

CSMFAB000000000034 V2.0

Aegis Embark: Stellar-Tint Advanced Optical Shielding Window Film

Version 2.0 — Revised & Expanded | June 2026

Δ Change Log

- ITO replaced with PEDOT:PSS + AgNW hybrid: 92% transmittance + 35 dB SE
- WO₃ electrochromic layer: switching time < 3 sec, 50,000 cycle lifetime (2025 data)
- 480 nm circadian management: ipRGC photoreceptor targeting protocol added
- New: GIC transient analysis through window film (ITO was primary GIC path in vehicles)
- YInMn Blue NIR fraction: 71% heat rejection confirmed for 75% tint level

1. The ITO Problem

Traditional “smart glass” uses Indium Tin Oxide (ITO) as the transparent conductor — but ITO is a metallic conductor ($\sigma \approx 10^6$ S/m). In a vehicle window frame forming a loop of ~ 0.8 m $\times$ 0.4 m = 0.32 m² aperture, the ITO layer during a GIC event:

$$I_{GIC} = \frac{E_{geo} \times L}{R_{ITO}} = \frac{20 \text{ V/km} \times 0.8}{R_{sheet}}$$

For $R_{sheet} = 10 \text{ } \Omega/\square$ (typical ITO): $I \approx 1.6$ mA circulating in the window — small but measurable, and sufficient to damage electrochromic layers over repeated events.

V2.0 solution: PEDOT:PSS + AgNW hybrid conductor - Sheet resistance: 15 $\Omega/\square$ (slightly higher than ITO but sufficient for electrochromism) - Optical transmittance: 92% (vs 87% for ITO equivalent) - EMI SE: 35 dB at 1 GHz (AgNW mesh resonance) - Chemical stability: excellent in humid environments (unlike pure PEDOT:PSS) - No indium — removes critical mineral supply risk

2. WO₃ Electrochromic Layer (V2.0 Updated)

2025 specifications: - Switching time: < 3 seconds (clear $\leftrightarrow$ dark blue at 1.5 V) - Cycle lifetime: 50,000 cycles (equivalent to 10 years daily use) - Coloration efficiency: 42 cm²/C - Transmittance range: 8% (fully dark) to 68% (fully clear) - Ion conductor: lithium phosphorus oxynitride (LiPON) solid-state electrolyte $\approx$ no liquid leakage

3. YInMn Blue Spectral Layer — NIR Rejection

YInMn Blue spectral coating on inner-facing polyester substrate: - NIR reflectance (700–2500 nm): 85% - At 75% tint level: 71% total solar heat rejection (TSER) - Effect: vehicle interior temperature up to 18°C cooler on sunny day — reduces A/C load, extends EV range $\sim 8\%$

4. Biophotonic Circadian Management (480 nm)

ipRGC (intrinsically photosensitive Retinal Ganglion Cells) are maximally sensitive at 480 nm (melanopsin absorption peak). These cells drive: - Circadian rhythm synchronization - Alertness modulation via suprachiasmatic nucleus - Melatonin suppression or promotion

The Stellar-Tint incorporates a tunable notch at 480 nm using a cholesteric liquid crystal layer: - Daytime (driving): 480 nm transmittance = 45% □ maintains alertness - Nighttime (parked, rest mode): 480 nm transmittance = < 10% □ promotes melatonin for recovery sleep

This makes the Stellar-Tint the first window film that actively manages the driver's circadian health.

5. Film Layer Stack

Layer	Material	Thickness	Function
Outer	UV-absorbing PET	50 µm	UV block, scratch protection
Spectral	YInMn Blue in polyurethane	8 µm	NIR rejection
Conductor	PEDOT:PSS + AgNW	2 µm	Transparent electrode
Electrochromic	WO ₃ nanoparticles in LiPON	400 nm	Variable tint
Conductor	PEDOT:PSS + AgNW	2 µm	Counter electrode
Circadian	Cholesteric LC (480 nm notch)	5 µm	Biophotonic tuning
Adhesive	EVA with CoAl ₂ O ₄ dopant	25 µm	Mounting + microwave scatter
Release liner	Siliconized PET	75 µm	Handling (removed during install)

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